

Bangladesh Army University of Science and Technology (BAUST), Saidpur

Department of Computer Science and Engineering (CSE)



PROPOSAL REPORT

Course Code: CSE 3200

Course Title: Web Engineering Project

Project Title: Tuition Management System (Study Bridge)

Comment:

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1. Introduction

1.1 Introduction

In Bangladesh, private tutoring is a very common and essential part of the education system. Students from school to university level often depend on tutors for better academic performance. However, finding a suitable tutor or tuition is still a manual and time-consuming process. Most students and guardians rely on Facebook groups, local agents, posters, or personal references.

This project aims to develop a Tuition Management System (Study Bridge) where students, guardians, and tutors from all over Bangladesh can connect through a secure, structured, and user-friendly digital platform. The system will automate tutor searching, profile management, tuition posting, and communication between tutors and students.

1.2 Motivation

Finding a compatible tutor is often a stressful race against time for parents, while many talented students and professionals struggle to find tuition opportunities that match their expertise. The motivation for this project is to democratize education by creating a centralized, searchable database. By solving this, we can ensure that a student in a remote area can find specialized help, and tutors can build a professional profile to secure a stable income.

1.3 Objectives

- ❖ To develop a user-friendly web application for real-time tutor-student matching.
- ❖ To implement a location-based search filter.
- ❖ To provide a rating and review system to ensure the quality and accountability of tutors.
- ❖ To create a secure dashboard for managing tuition applications and status updates.

1.4 Feasibility Study

Existing solutions in Bangladesh include Facebook groups and a few localized websites. While Facebook groups have high engagement, they lack structured data and are prone to scams. Established platforms often focus heavily on Dhaka or big cities, leaving other regions underserved. Research into existing systems shows a move toward Uber-like models for services, but education requires a more detailed profile-matching algorithm to be effective.

1.5 Gap Analysis

Most current platforms act merely as bulletin boards. There is a significant gap in verified credentials, geographic reach, and automated matching. This project intends to fill that gap

by introducing a verification badge system and a robust backend that handles tutoring requests systematically rather than just displaying phone numbers.

2. Proposed System Architecture

2.1 Requirement Analysis & Design Specification

2.1.1 Overview

The system will follow a Model-View-Controller (MVC) architecture. The front end will be responsive to ensure accessibility via web, while the backend will handle complex queries for location and subject filtering.

2.1.2 Proposed System Design

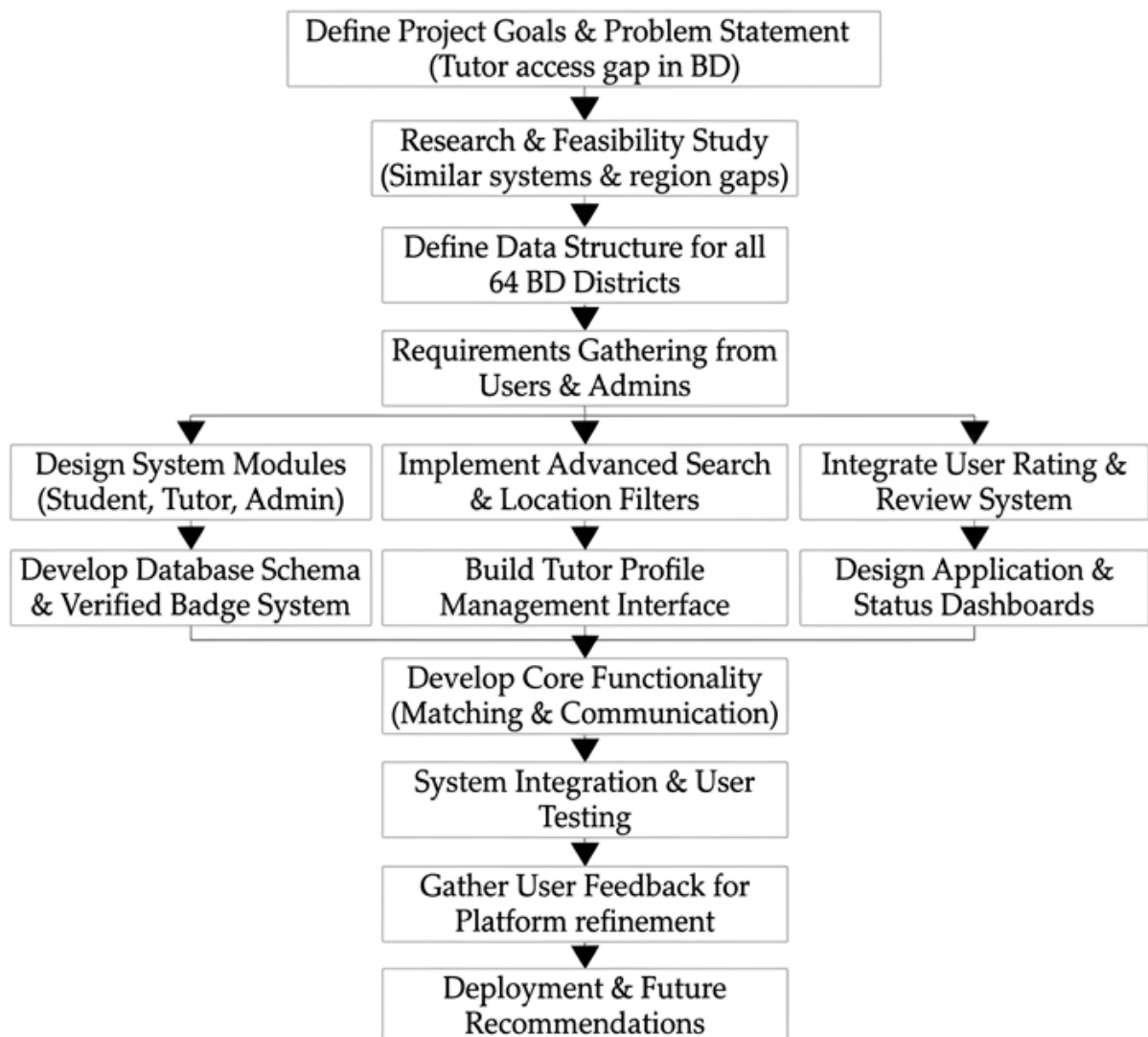


Fig-01: Methodological Flowchart

2.2 Overall Project Plan

The development is divided into four focused phases to ensure a structured delivery of the Study Bridge.

Phase 1: Analysis & Schema: Requirement finalization and Database. Designing the UI/UX.

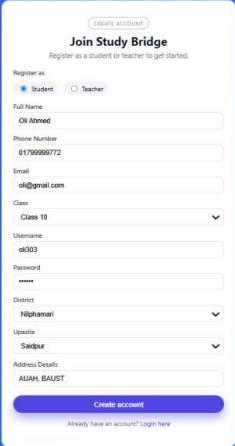
Phase 2: Core Development: Setting up Node.js server and PostgreSQL database.

Phase 3: Integration & UI: Developing the React.js frontend with Tailwind CSS. Connecting the Tutor/Student dashboards to the backend API.

Phase 4: Testing & Deployment: Bug fixing, web responsiveness testing, and security checks. Launching the live website on Vergel/Render.

3. Implementation

3.1 UI Design



CREATE ACCOUNT

Join Study Bridge

Register as a student or teacher to get started.

Register as: ☒ Student ☐ Teacher

Full Name:

Phone Number:

Email:

Class:

Username:

Password:

Classics:

Upstate:

Address Details:

Create account

Already have an account? Login here

Fig-02: Study Bridge Register page (Demo)

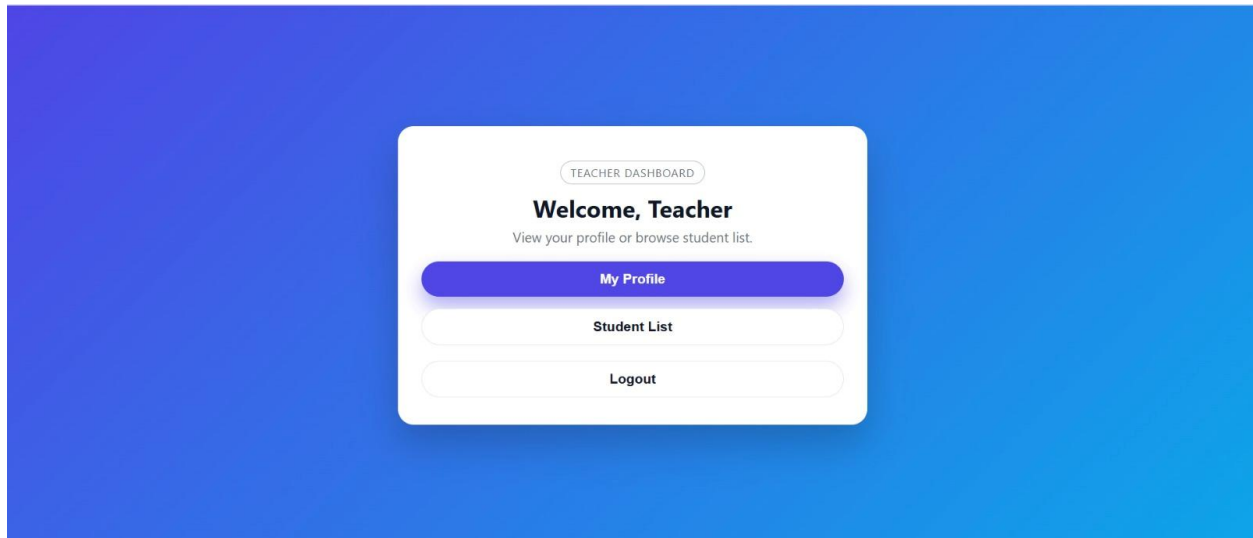


Fig-03: Study Bridge Teacher List page (Demo)

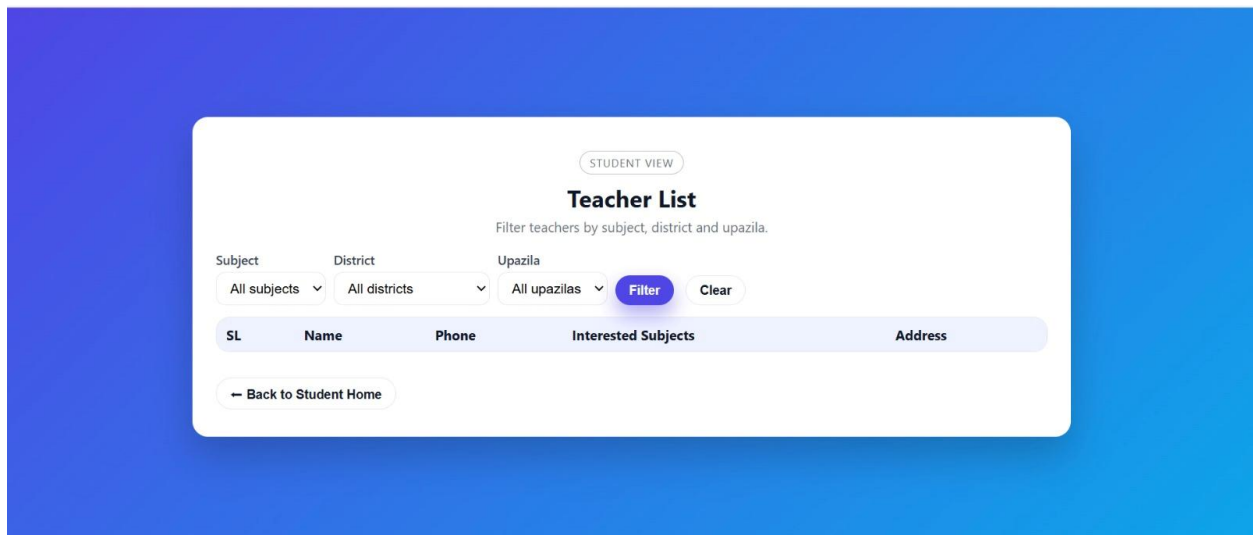


Fig-04: Study Bridge Welcome page (Demo)

4. Conclusion

4.1 Summary

The Study Bridge provides a streamlined, digital alternative to the chaotic traditional tutoring market in Bangladesh. By centralizing data and providing search tools, it saves time for parents and creates opportunities for educators.